# NUMBER STATISTICS

## 1. Introduction

Number Statistics is a Python-based project that analyzes a list of integers. The program identifies even and odd numbers, separates them into two lists, and finds the largest and smallest numbers.

This project demonstrates how Python can be used to perform basic numerical processing through loops, conditional statements, and built-in functions.

## 2. Objective

The objectives of this project are:

- To identify even and odd numbers in a list.
- To create separate lists for even and odd values.
- To find the largest and smallest numbers.
- To practise loops and conditional statements.
- To understand the use of the modulo operator in numerical processing.

## 3. Problem Statement

When working with a list of numbers, manually classifying even and odd values and finding the largest and smallest values can be repetitive.

This project automates the process using Python and presents the results in a clear and structured format.

## 4. Tools and Technologies Used

**Python 3:** Used to implement the number classification and statistical operations.

**Visual Studio Code:** Used to write, edit, and execute the Python program.

**GitHub:** Used to host the source code and project documentation.

## 5. Dataset Description

The project uses a manually created list of eight integers:

`[12, 7, 18, 5, 0, -3, 22, 9]`

The dataset contains positive numbers, negative numbers, zero, even numbers, and odd numbers. This allows the program to demonstrate different classification cases.

## 6. Methodology

The project follows these steps:

1. Create a list containing integers.
2. Check that the list is not empty.
3. Validate that each value is an integer.
4. Process each number using a loop.
5. Use the modulo operator to determine whether each number is even or odd.
6. Append even numbers to the even list and odd numbers to the odd list.
7. Use the `max()` function to find the largest number.
8. Use the `min()` function to find the smallest number.
9. Display the original list and the calculated results.

## 7. Concepts and Logic

### Even and Odd Numbers

An integer is even if it is divisible by 2 without a remainder. Otherwise, it is odd.

The modulo operator (`%`) returns the remainder after division.

For example:

- `12 % 2 == 0`, so 12 is even.
- `7 % 2 == 1`, so 7 is odd.
- `0 % 2 == 0`, so zero is even.
- `-3 % 2 != 0`, so -3 is odd.

### Largest and Smallest Numbers

The built-in `max()` function returns the largest value in a non-empty list.

The built-in `min()` function returns the smallest value in a non-empty list.

## 8. Results and Output

For the sample dataset, the expected results are:

**Original numbers:** `[12, 7, 18, 5, 0, -3, 22, 9]`

**Even numbers:** `[12, 18, 0, 22]`

**Odd numbers:** `[7, 5, -3, 9]`

**Largest number:** 22

**Smallest number:** -3

These results should be verified by executing the Python program.

**Screenshot Evidence:** Insert a screenshot of the actual program output here.

## 9. Testing and Validation

The program should be tested with different inputs, including:

- A list containing both even and odd numbers.
- A list containing only even numbers.
- A list containing only odd numbers.
- Negative numbers and zero.
- A list containing a single integer.
- A list containing repeated values.
- An empty list.
- Non-integer values.

The program should classify valid integers correctly and display errors for empty lists or invalid input values.

Actual test outcomes should be recorded after execution.

## 10. Concepts Learned

This project provides practical experience with:

- Python lists and variables.
- `for` loops.
- `if-else` statements.
- The modulo operator.
- List methods such as `append()`.
- Built-in functions such as `max()` and `min()`.
- Input validation and exception handling.
- Structured output formatting.

## 11. Limitations and Future Improvements

The current implementation uses a small, manually created dataset. The numbers are not entered interactively by the user.

Future improvements could include:

- Accepting numbers from user input.
- Reading numbers from a text or CSV file.
- Calculating the sum and average.
- Counting even and odd numbers.
- Displaying a visual representation of the results.
- Creating an interactive interface.

## 12. Conclusion

The Number Statistics project demonstrates how Python can process a list of integers and classify values as even or odd. It also identifies the largest and

smallest numbers using built-in functions.

The project strengthens fundamental Python programming skills and provides a foundation for more advanced numerical data analysis.

## 13. Project Links

GitHub Repository: https://github.com/venky1318/number_statistics